Proof by Contradiction and an Introduction to Functions

Name: _____

Part I. Proof by contradiction — 56 minutes (70%)

For each problem, write a proof by contradiction. State your assumption, identify the contradiction explicitly, and conclude the original claim.

Recall. To prove a statement P by contradiction, assume $\neg P$ and derive a contradiction. To prove $P \Rightarrow Q$ this way, assume *both* P and $\neg Q$.

1. Sets

Let A and B be sets. Prove that

$$A \cap (B \setminus A) = \emptyset.$$

If the intersection were nonempty, what element could you choose? Explain which two membership statements would contradict each other.

2. There is no largest even integer

Prove that there is no largest even integer.

Begin by stating precisely what it would mean for an even integer m to be the largest. Construct an even integer that contradicts that assumption.

Part I. Proof by contradiction (continued)

3. Even squares

Let $a \in \mathbb{Z}$. Prove that if a^2 is even, then a is even.

First write down the hypothesis together with the negation of the conclusion. You may use the fact that every integer is either even or odd, but not both.

4. Rational and irrational numbers

Let $a, b \in \mathbb{R}$. Prove that if a is rational and ab is irrational, then b is irrational.

Use the definition of a rational number. When you introduce fractions, state which quantities are integers and which denominators are nonzero.

Check your proof. Underline the assumption you made for contradiction and circle the two statements that cannot both be true.

Part I. Putting the arguments together

5. An impossible equation

Prove that for every $a, b \in \mathbb{Z}$,

$$a^2 - 4b \neq 2.$$

You may use the result of Problem 3. If the equation $a^2 - 4b = 2$ held, what would it tell you about the parity of a^2, and then of a?

6. A sum of two squares

Let $a, b, c \in \mathbb{Z}$. Prove that if $a^2 + b^2 = c^2$, then at least one of a or b is even.

Negate "at least one is even" carefully. You may use Problems 3 and 5.

Before moving on. In a proof of $P \Rightarrow Q$ by contradiction, why must you retain the hypothesis P when you assume $\neg Q$?

Part II. Functions 24 minutes (30%)

Definitions. A function $f\colon X \to Y$ assigns to each $x \in X$ exactly one value $f(x) \in Y$. Its **domain** is X, its **codomain** is Y, and its **image** is $f(X) = \{f(x) : x \in X\}$.

- **Injective (one-to-one):** for all $x_1, x_2 \in X$, $f(x_1) = f(x_2) \Rightarrow x_1 = x_2$.
- **Surjective (onto):** for every $y \in Y$, there is an $x \in X$ with $f(x) = y$.

7. The same formula with different domains and codomains

For each assignment, decide whether it defines a function. If it does, give its image and decide whether it is injective and whether it is surjective. If it does not, give an input that shows the failure and mark the last three columns N/A. Give brief explanations below the table.

Assignment	Function?	Image	Injective?	Surjective?
$f\colon \mathbb{R} \to \mathbb{R}$, $f(x) = x/5$				
$g\colon \mathbb{Z} \to \mathbb{Z}$, $g(x) = x/5$				
$h\colon \mathbb{Q} \to \mathbb{R}$, $h(x) = x/5$				
$\ell\colon \mathbb{R} \to \mathbb{Q}$, $\ell(x) = x/5$				

Explanations: For an injective function, explain why equal outputs force equal inputs. For a surjective function, show how to find an input for an arbitrary target; for a nonsurjective function, identify a target outside its image.

Discuss. How can the formula $x/5$ define a function in one row but fail to define a function in another? What is the difference between the image and the codomain?

Part II. Injectivity and surjectivity (continued)

8. Disproving injectivity

Prove or disprove: $f\colon \mathbb{R} \to \mathbb{R}$, defined by $f(x) = x^3 - x$, is injective. If you disprove it, exhibit two distinct inputs with the same output.

9. Proving surjectivity

Prove that $g\colon \mathbb{R}^2 \to \mathbb{R}^2$, defined by $g(x, y) = (x + y, x - y)$, is surjective.

Start with an arbitrary target $(u, v) \in \mathbb{R}^2$. Find real numbers x, y satisfying $x + y = u$ and $x - y = v$, and check your answer.